

# A Fractal Representation of the Complex Clifford Algebra Equivalent to the Fock Representation

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**Abstract.** We construct a representation of the infinite dimensional complex Clifford algebra on the Hilbert space of square-integrable complex-valued functions on the Cantor set, which we show to be equivalent to the classical Fock representation.

**Keywords.** Fock representation, Clifford algebra, Cantor set, fractal representation.

## 1. Introduction

It is known that the (countable-)infinite dimensional complex Clifford algebra can be represented on the Hilbert space which is the completion of the exterior product space of an infinite dimensional Hilbert space. This representation is called the Fock representation [1]. In [2] we have constructed representations of real and complex Clifford algebras on function spaces on the Cantor set, which have a more geometric flavour than the Fock representation. In the present note, we construct a representation of the infinite dimensional complex Clifford algebra on the Hilbert space of square-integrable complex-valued functions on the Cantor set along the lines of [2], which is devised to yield an equivalence with the Fock representation. We refer for the relevant definitions to [2] and recall only the following most important lemma:

**Lemma 1.1.** *There are bounded, linear operators  $T_p, S_p$  ( $p \in \mathbb{N}$ ) on the complex Hilbert space  $L^2\mathcal{K}$  of square-integrable, complex-valued functions on the Cantor set  $\mathcal{K}$  (with respect to the  $\ln 2/\ln 3$ -dimensional Hausdorff measure) satisfying the following commutation rules:*

- i)  $T_p^2 = S_p^2 = Id$  ( $p \in \mathbb{N}$ )
- ii)  $T_p T_q = T_q T_p$  ( $p, q \in \mathbb{N}$ )
- iii)  $S_p S_q = S_q S_p$  ( $p, q \in \mathbb{N}$ )
- iv)  $T_p S_q = S_q T_p$  ( $p, q \in \mathbb{N}, p \neq q$ )
- v)  $T_p S_p = -S_p T_p$  ( $p \in \mathbb{N}$ ).

As usual, a representation of the infinite dimensional complex Clifford algebra  $\mathbb{Cl}_\infty$  on  $L^2\mathcal{K}$  will be an algebra homomorphism  $\mathbb{Cl}_\infty \rightarrow B(L^2\mathcal{K})$ , where  $B(L^2\mathcal{K})$  is the  $C^*$ -algebra of bounded (complex)linear operators on  $L^2\mathcal{K}$ .

## 2. A Representation of the Infinite Dimensional Complex Clifford Algebra on $L^2\mathcal{K}$

Let us denote the generators of the infinite dimensional complex Clifford algebra  $\mathbb{Cl}_\infty$  by  $e_j$  (with  $e_j^2 = 1$  and  $e_j e_k = -e_k e_j$  for  $j \neq k$ ,  $j, k = 1, 2, \dots$ ). Let us map these generators in the following way into  $B(L^2\mathcal{K})$ :

$$\begin{aligned}
 \psi : \quad e_1 &\longmapsto T_1 \\
 e_2 &\longmapsto iT_1 S_1 \\
 e_3 &\longmapsto T_2 S_1 \\
 e_4 &\longmapsto iT_2 S_2 S_1 \\
 &\vdots \\
 (k > 1) \quad e_{2k-1} &\longmapsto T_k S_{k-1} S_{k-2} \cdots S_2 S_1 \\
 e_{2k} &\longmapsto iT_k S_k S_{k-1} \cdots S_2 S_1 \\
 &\vdots
 \end{aligned} \tag{2.1}$$

( $i$  is the complex  $i$ .)

**Theorem 2.1.**  $\psi$  induces an algebra homomorphism  $\mathbb{Cl}_\infty \rightarrow B(L^2\mathcal{K})$ , i.e. a representation of  $\mathbb{Cl}_\infty$  on  $L^2\mathcal{K}$ .

*Proof.* We have to see,

$$(\psi(e_p))^2 = I$$

and

$$\psi(e_p)\psi(e_q) = -\psi(e_q)\psi(e_p)$$

for  $p, q \in \{1, 2, \dots\}$  and  $p \neq q$ .

- Let us first check that  $(\psi(e_p))^2 = I$ . Obviously,

$$(\psi(e_1))^2 = T_1 T_1 = I$$

and

$$(\psi(e_2))^2 = (iT_1 S_1)(iT_1 S_1) = -T_1 S_1 T_1 S_1 = T_1 T_1 S_1 S_1 = I.$$

Now let  $p = 2k - 1$  ( $k > 1$ ):

$$\begin{aligned}
 (\psi(e_{2k-1}))^2 &= (T_k S_{k-1} S_{k-2} \cdots S_2 S_1)(T_k S_{k-1} S_{k-2} \cdots S_2 S_1) \\
 &= (T_k T_k S_{k-1} S_{k-1} S_{k-2} S_{k-2} \cdots S_2 S_2 S_1 S_1) \text{ (by Lemma 1.1)} \\
 &= I.
 \end{aligned}$$

For  $p = 2k$  we get

$$\begin{aligned}
 (\psi(e_{2k}))^2 &= (iT_k S_k S_{k-1} \cdots S_2 S_1)(iT_k S_k S_{k-1} \cdots S_2 S_1) \\
 &= -i^2 (T_k T_k S_k S_k S_{k-1} S_{k-1} \cdots S_2 S_2 S_1 S_1) \\
 &= I.
 \end{aligned}$$

- Let us now check the anti-commutativity relations. By Lemma 1.1,

$$\psi(e_1)\psi(e_2) = -\psi(e_2)\psi(e_1).$$

Likewise,  $\psi(e_1)$  and  $\psi(e_2)$  anti-commute with all  $\psi(e_j)$  for  $j > 2$ .

Now we consider various cases:

- i) Let  $p = 2k - 1$ ,  $q = 2l - 1$  for  $k > 1, l > 1$  and  $k < l$ .

$$\begin{aligned}
 \psi(e_p)\psi(e_q) &= (T_k S_{k-1} S_{k-2} \cdots S_2 S_1)(T_l S_{l-1} S_{l-2} \cdots S_2 S_1) \\
 &= (T_k)(T_l S_{l-1} \cdots S_k \cdots S_1)(S_{k-1} S_{k-2} \cdots S_2 S_1) \\
 &= -(T_l S_{l-1} \cdots S_2 S_1)(T_k S_{k-1} S_{k-2} \cdots S_2 S_1) \\
 &= -\psi(e_q)\psi(e_p).
 \end{aligned}$$

- ii) Let  $p = 2k$ ,  $q = 2l$  for  $k > 1, l > 1$  and  $k < l$ .

$$\begin{aligned}
 \psi(e_p)\psi(e_q) &= (iT_k S_k S_{k-1} \cdots S_2 S_1)(iT_l S_l S_{l-1} \cdots S_2 S_1) \\
 &= (iT_k)(iT_l S_l S_{l-1} \cdots S_k \cdots S_1)(S_k S_{k-1} \cdots S_1) \\
 &= -(iT_l S_l S_{l-1} \cdots S_2 S_1)(iT_k S_k S_{k-1} \cdots S_1) \\
 &= -\psi(e_q)\psi(e_p).
 \end{aligned}$$

- iii) Let  $p = 2k - 1 < q = 2l$ ,  $k > 1, l > 1$ .

$$\begin{aligned}
 \psi(e_p)\psi(e_q) &= (T_k S_{k-1} S_{k-2} \cdots S_2 S_1)(iT_l S_l S_{l-1} \cdots T_1 S_1) \\
 &= (T_k)(iT_l S_l S_{l-1} \cdots S_k \cdots S_1)(S_{k-1} S_{k-2} \cdots S_1) \\
 &= -(iT_l S_l S_{l-1} \cdots S_1)(T_k S_{k-1} S_{k-2} \cdots S_1) \\
 &= -\psi(e_q)\psi(e_p).
 \end{aligned}$$

The remaining cases are seen similarly.  $\square$

### 3. Fock Representations for the Infinite Dimensional Complex Clifford Algebra

In this section, we take a real Hilbert space  $(V, (\cdot, \cdot))$  and recall the Fock representations of its (plain)complex Clifford algebra  $C(V)$ . To define the Fock representation it is required that  $V$  be converted into a complex Hilbert space  $V_J$  by a unitary structure  $J$  which is an orthogonal transformation of  $V$  whose square is minus the identity. The assumption,  $J^2 = -I$ , allows us to convert  $V$  into a complex vector space with  $iv = J(v)$  for all  $v \in V$ . We

will write  $V_J$  for the vector space  $V$  made complex via  $J$ . Given  $x, y \in V$  we shall put

$$\langle x, y \rangle_J = (x, y) + i(x, J(y)).$$

This defines a Hermitian inner product  $\langle \cdot, \cdot \rangle_J$  on the complex vector space  $V_J$ . For positive  $n$ , the  $n$ -fold alternating power  $\bigwedge^n(V_J)$  carries a canonical Hermitian inner product with

$$\langle x_1 \wedge \cdots \wedge x_n, y_1 \wedge \cdots \wedge y_n \rangle = \text{Det}[\langle x_i, y_j \rangle_J]$$

for  $x_1, y_1, \dots, x_n, y_n \in V$ . We note that  $\bigwedge^0(V_J) = \mathbb{C}$ . The exterior product space,

$$H_J(V) = \bigwedge(V_J) = \bigoplus_{n \geq 0} \bigwedge^n(V_J),$$

has elements  $\bigoplus_{n \geq 0} \zeta_n$  where  $\zeta_n \in \bigwedge^n(V_J)$  is zero for all but finitely many values of  $n \geq 0$ . The inner product is given by

$$\langle \bigoplus_{n \geq 0} \xi_n, \bigoplus_{n \geq 0} \zeta_n \rangle = \sum_{n \geq 0} \langle \xi_n, \zeta_n \rangle$$

where  $\bigoplus_{n \geq 0} \xi_n, \bigoplus_{n \geq 0} \zeta_n \in H_J(V)$ . The Hilbert space completion of  $H_J(V)$  is the Fock space  $\mathbb{H}_J(V)$  over  $V$  determined by the unitary structure  $J$ .

Now we define the creator and annihilator on the exterior algebra  $\bigwedge(V_J) = H_J(V)$  over  $V_J$  to be used to define the Fock representation. If  $v \in V$  then the creator (or creation operator)  $c(v) = c_J(v)$  is the complex-linear operator on  $H_J(V)$  given by

$$\begin{array}{ccc} c(v) : & H_J(V) & \rightarrow H_J(V) \\ & \zeta & \mapsto v \wedge \zeta. \end{array}$$

The annihilator (or annihilation operator)  $a(v) = a_J(v)$  is the complex-linear operator on  $H_J(V)$  given on decomposables by the rule

$$a(v)(w_0 \wedge \cdots \wedge w_n) = \sum_{j=0}^n (-1)^j \langle w_j, v \rangle_J w_0 \wedge \cdots \wedge \hat{w}_j \wedge \cdots \wedge w_n$$

for  $w_0, \dots, w_n \in V$ .

**Theorem 3.1.** *If  $v \in V$ , then  $c(v)$  and  $a(v)$  extend continuously to mutually adjoint operators of norm  $\|v\|$  on  $\mathbb{H}_J(V)$ .*

*Proof.* See [1]. □

The Fock representation will take place on the Fock space  $\mathbb{H}_J(V)$  and will be defined in terms of creators and annihilators.

**Definition 3.2.** The map  $\pi_J : V \rightarrow B(\mathbb{H}_J(V))$

$$v \mapsto c_J(v) + a_J(v)$$

satisfies  $(\pi_J(v))^2 = 1$  and thus defines a representation  $\pi_J$  of the real Clifford algebra  $C_{\mathbb{R}}(V)$  (with  $v^2 = (v, v).1$ ):

$$\pi_J : C_{\mathbb{R}}(V) \rightarrow B(\mathbb{H}_J(V)).$$

(We used the same symbol again.) This extends to the (plain) complex Clifford algebra

$$C(V) \cong C_{\mathbb{R}}(V) \otimes \mathbb{C}$$

by  $u \otimes z \mapsto z\pi_J(u)$ , for  $z \in \mathbb{C}$ . The representation of  $C(V)$  defined in this way is called the Fock representation  $\pi_J$  of the (plain) complex Clifford algebra  $C(V)$ :

$$\begin{aligned} \pi_J : C(V) &\longrightarrow B(\mathbb{H}_J(V)) \\ v &\longmapsto c_J(v) + a_J(v), \text{ for } v \in V. \end{aligned}$$

Let  $V$  be an infinite dimensional real Hilbert space with the orthonormal basis  $\{e_1, e_2, \dots\}$  and  $J$  the associated standard unitary structure:

$$\begin{aligned} J : V &\longrightarrow V \\ e_1 &\longmapsto e_2 \\ e_2 &\longmapsto -e_1 \\ &\vdots \\ (k > 1) \quad e_{2k-1} &\longmapsto e_{2k} \\ e_{2k} &\longmapsto -e_{2k-1} \\ &\vdots \end{aligned} \tag{3.1}$$

We can now formulate our main result:

**Theorem 3.3.** *The representation  $\psi$  (given by (2.1)) of  $\mathbb{C}l_{\infty}$  on  $L^2\mathcal{K}$  is equivalent to the Fock representation  $\pi_J$  of  $C(V) = \mathbb{C}l_{\infty}$  on the Fock space  $\mathbb{H}_J(V)$ , where the unitary structure is given by (3.1).*

*Proof.* We must find an isomorphism  $f$  between the spaces  $L^2\mathcal{K}$  and the Fock space  $\mathbb{H}_J(V)$  such that the following diagram commutes for all  $e_i$ ,  $i \geq 1$ .

$$\begin{array}{ccc} L^2\mathcal{K} & \xrightarrow{\psi(e_i)} & L^2(\mathcal{K}) \\ f \downarrow & & \downarrow f \\ \mathbb{H}_J(V) & \xrightarrow{\pi_J(e_i)} & \mathbb{H}_J(V) \end{array} \tag{3.2}$$

Let's take the orthonormal basis

$$\{1, T_1(1), T_2(1), T_1T_2(1), T_3(1), T_1T_3(1), T_2T_3(1), T_1T_2T_3(1), \dots\}$$

for  $L^2\mathcal{K}$  (see [3]) and

$$\{1, e_1, e_3, e_1 \wedge e_3, e_5, e_1 \wedge e_5, e_3 \wedge e_5, e_1 \wedge e_3 \wedge e_5, \dots\}$$

for  $\mathbb{H}_J(V)$  (see [1]). Now we define  $f$  by

$$\begin{array}{rcl}
 f : L^2\mathcal{K} & \longrightarrow & \mathbb{H}_J(V) \\
 1 & \longmapsto & 1 \\
 T_1(1) & \longmapsto & e_1 \\
 T_2(1) & \longmapsto & e_3 \\
 T_1T_2(1) & \longmapsto & e_1 \wedge e_3 \\
 T_3(1) & \longmapsto & e_5 \\
 T_1T_3(1) & \longmapsto & e_1 \wedge e_5 \\
 T_2T_3(1) & \longmapsto & e_3 \wedge e_5 \\
 T_1T_2T_3(1) & \longmapsto & e_1 \wedge e_3 \wedge e_5 \\
 & \vdots & \\
 T_k(1) & \longmapsto & e_{2k-1} \\
 i_1 < \dots < i_r, \quad T_{i_1}T_{i_2} \dots T_{i_r}(1) & \longmapsto & e_{2i_1-1} \wedge e_{2i_2-1} \wedge \dots \wedge e_{2i_r-1} \\
 & \vdots &
 \end{array}$$

We have to show that the following element-diagram commutes for any base element  $T_{i_1} \dots T_{i_k}(1) \in L^2\mathcal{K}$  and any generator  $e_i \in \mathbb{C}l_\infty$  :

$$\begin{array}{ccc}
 T_{i_1} \dots T_{i_k}(1) & \xrightarrow{\psi(e_i)} & * \\
 f \downarrow & & \downarrow f \\
 * & \xrightarrow{\pi_J(e_i)} & *
 \end{array}$$

As a warm-up, let us check that the first few element-diagrams are commutative:

$$\begin{array}{ll}
 \text{a)} \quad \begin{array}{ccc} 1 & \xrightarrow{\psi(e_1)} & T_1(1) \\ f \downarrow & & \downarrow f \\ 1 & \xrightarrow{\pi_J(e_1)} & e_1 \end{array} & \text{b)} \quad \begin{array}{ccc} T_1(1) & \xrightarrow{\psi(e_1)} & 1 \\ f \downarrow & & \downarrow f \\ e_1 & \xrightarrow{\pi_J(e_1)} & 1 \end{array} \\
 \text{c)} \quad \begin{array}{ccc} T_2(1) & \xrightarrow{\psi(e_1)} & T_1T_2(1) \\ f \downarrow & & \downarrow f \\ e_3 & \xrightarrow{\pi_J(e_1)} & e_1 \wedge e_3 \end{array} & \text{d)} \quad \begin{array}{ccc} T_1T_2(1) & \xrightarrow{\psi(e_1)} & T_2(1) \\ f \downarrow & & \downarrow f \\ e_1 \wedge e_3 & \xrightarrow{\pi_J(e_1)} & e_3 \end{array}
 \end{array}$$

a)

$$(f \circ \psi(e_1))(1) = f(T_1(1)) = e_1.$$

On the other hand,

$$\begin{aligned}
 (\pi_J(e_1) \circ f)(1) &= \pi_J(e_1)(1) = c_J(e_1)(1) + a_J(e_1)(1) \\
 &= e_1 \wedge 1 \\
 &= e_1.
 \end{aligned}$$

Thus we get the equality,

$$(f \circ \psi(e_1))(1) = (\pi_J(e_1) \circ f)(1).$$

b)

$$(f \circ \psi(e_1))(T_1(1)) = f(T_1 T_1(1)) = f(1) = 1.$$

Besides,

$$\begin{aligned} (\pi_J(e_1) \circ f)(T_1(1)) &= \pi_J(e_1)(e_1) \\ &= c_J(e_1)(e_1) + a_J(e_1)(e_1) \\ &= 1. \end{aligned}$$

Thus we get the equality,

$$(f \circ \psi(e_1))(T_1(1)) = (\pi_J(e_1) \circ f)(T_1(1)).$$

c)

$$(f \circ \psi(e_1))(T_2(1)) = f(\psi(e_1)(T_2(1))) = f(T_1 T_2(1)) = e_1 \wedge e_3.$$

However we have also the equality,

$$\begin{aligned} (\pi_J(e_1) \circ f)(T_2(1)) &= \pi_J(e_1)(f(T_2(1))) \\ &= \pi_J(e_1)(e_3) \\ &= c_J(e_1)(e_3) + a_J(e_1)(e_3) \\ &= e_1 \wedge e_3 + e_3, e_1 >_J \\ &= e_1 \wedge e_3. \end{aligned}$$

Thus,

$$(f \circ \psi(e_1))(T_2(1)) = (\pi_J(e_1) \circ f)(T_2(1)).$$

d)

$$(f \circ \psi(e_1))(T_1 T_2(1)) = f(T_1 T_1 T_2(1)) = f(T_2(1)) = e_3.$$

On the other hand,

$$\begin{aligned} (\pi_J(e_1) \circ f)(T_1 T_2(1)) &= \pi_J(e_1)(f(T_1 T_2(1))) \\ &= \pi_J(e_1)(e_1 \wedge e_3) \\ &= c_J(e_1)(e_1 \wedge e_3) + a_J(e_1)(e_1 \wedge e_3) \\ &= e_1 \wedge e_1 \wedge e_3 + e_1, e_1 >_J e_3 - e_3, e_1 >_J e_1 \\ &= e_3. \end{aligned}$$

Thus we get,

$$(f \circ \psi(e_1))(T_1 T_2(1)) = (\pi_J(e_1) \circ f)(T_1 T_2(1)).$$

Now we will show the general case for any generator element  $e_i \in \mathcal{Cl}_\infty$  and any base element  $T_{i_1} T_{i_2} \cdots T_{i_k}(1) \in L^2 K$  :

$$(f \circ \psi(e_i))(T_{i_1} T_{i_2} \cdots T_{i_k}(1)) = (\pi_J(e_i) \circ f)(T_{i_1} T_{i_2} \cdots T_{i_k}(1)).$$

We consider two cases:  $i = 2l - 1$  and  $i = 2l$ .

- Let  $i = 2l - 1$  ( $l \geq 1$ ):

$$\begin{aligned} (f \circ \psi(e_{2l-1}))(T_{i_1} T_{i_2} \cdots T_{i_k}(1)) &= f(\psi(e_{2l-1})(T_{i_1} T_{i_2} \cdots T_{i_k}(1))) \\ &= f(T_l S_{l-1} S_{l-2} \cdots S_2 S_1 T_{i_1} T_{i_2} \cdots T_{i_k}(1)). \end{aligned}$$

Let's define the set  $A$  as

$$A = \{i_j \mid \exists t = 1, 2, \dots, l-1, i_j = t, j = 1, 2, \dots, k\}.$$

Then,

$$S_{l-1} S_{l-2} \cdots S_2 S_1 T_{i_1} T_{i_2} \cdots T_{i_k}(1) = (-1)^{|A|} T_{i_1} T_{i_2} \cdots T_{i_k}(1)$$

where  $|A|$  is the cardinality of  $A$ .

$$\begin{aligned} &f(\psi(e_{2l-1})(T_{i_1} \cdots T_{i_k}(1))) = f((-1)^{|A|} T_{i_1} T_{i_2} \cdots T_{i_k}(1)) \\ &= \begin{cases} f((-1)^{j_0-1} T_{i_1} \cdots \hat{T}_{i_{j_0}} \cdots T_{i_k}(1)) & , \quad i_{j_0} = l \\ f((-1)^{|A|} T_l T_{i_1} \cdots T_{i_k}(1)) & , \quad j = 1, \dots, k, i_j \neq l \end{cases} \\ &= \begin{cases} (-1)^{j_0-1} e_{2i_1-1} \wedge \cdots \wedge \hat{e}_{2i_{j_0}-1} \wedge \cdots \wedge e_{2i_k-1} , & i_{j_0} = l \\ e_{2l-1} \wedge e_{2i_1-1} \wedge \cdots \wedge e_{2i_k-1} , & 1 \leq j \leq k, l \leq i_j \quad (|A| = 0) \\ (-1)^{j_1} e_{2i_1-1} \wedge \cdots \wedge e_{2i_{j_1}-1} \wedge e_{2l-1} \wedge e_{2i_{j_2}-1} \\ \quad \wedge \cdots \wedge e_{2i_k-1} , & i_{j_1} < l < i_{j_2} \quad (|A| = j_1) \\ (-1)^k e_{2i_1-1} \wedge \cdots \wedge e_{2i_k-1} \wedge e_{2l-1} , & 1 \leq j \leq k, l \geq i_j \quad (|A| = k). \end{cases} \end{aligned} \quad (3.3)$$

On the other hand,

$$\begin{aligned} \pi_J(e_{2l-1})[f(T_{i_1} T_{i_2} \cdots T_{i_k}(1))] &= \pi_J(e_{2l-1})[e_{2i_1-1} \wedge e_{2i_2-1} \wedge \cdots \wedge e_{2i_k-1}] \\ &= \begin{cases} (-1)^{j_0-1} e_{2i_1-1} \wedge \cdots \wedge \hat{e}_{2i_{j_0}-1} \wedge \cdots \wedge e_{2i_k-1} , & i_{j_0} = l \\ e_{2l-1} \wedge e_{2i_1-1} \wedge \cdots \wedge e_{2i_k-1} , & 1 \leq j \leq k, l \leq i_j \\ (-1)^{j_1} e_{2i_1-1} \wedge \cdots \wedge e_{2i_{j_1}-1} \wedge e_{2l-1} \wedge e_{2i_{j_2}-1} \wedge \cdots \wedge e_{2i_k-1} , & i_{j_1} < l < i_{j_2} \\ (-1)^k e_{2i_1-1} \wedge \cdots \wedge e_{2i_k-1} \wedge e_{2l-1} , & 1 \leq j \leq k, l \geq i_j. \end{cases} \end{aligned} \quad (3.4)$$

From the equalities (3.3) and (3.4) we obtain,

$$f(\psi(e_{2l-1})(T_{i_1} T_{i_2} \cdots T_{i_k}(1))) = \pi_J(e_{2l-1})[f(T_{i_1} T_{i_2} \cdots T_{i_k}(1))].$$

- Let  $i = 2l$  ( $l \geq 1$ ): We have to show

$$(f \circ \psi(e_{2l}))(T_{i_1} T_{i_2} \cdots T_{i_k}(1)) = (\pi_J(e_{2l}) \circ f)(T_{i_1} T_{i_2} \cdots T_{i_k}(1)).$$

$$\begin{aligned} (f \circ \psi(e_{2l}))(T_{i_1} T_{i_2} \cdots T_{i_k}(1)) &= f(\psi(e_{2l})(T_{i_1} T_{i_2} \cdots T_{i_k}(1))) \\ &= f(i T_l S_l S_{l-1} S_{l-2} \cdots S_2 S_1 T_{i_1} T_{i_2} \cdots T_{i_k}(1)). \end{aligned}$$

In this case let's define the set  $B$  as

$$B = \{i_j \mid \exists t = 1, 2, \dots, l, i_j = t, j = 1, 2, \dots, k\}.$$

Then the equality

$$S_l S_{l-1} S_{l-2} \cdots S_2 S_1 T_{i_1} T_{i_2} \cdots T_{i_k}(1) = (-1)^{|B|} T_{i_1} T_{i_2} \cdots T_{i_k}(1)$$



holds, where  $|B|$  is the cardinality of  $B$ .

$$\begin{aligned}
 f(\psi(e_{2l})(T_{i_1}T_{i_2}\cdots T_{i_k}(1))) &= f(iT_lS_lS_{l-1}S_{l-2}\cdots S_2S_1T_{i_1}T_{i_2}\cdots T_{i_k}(1)) \\
 &= f((-1)^{|B|}iT_lT_{i_1}T_{i_2}\cdots T_{i_k}(1)) \\
 &= \begin{cases} f(i(-1)^{|B|}T_{i_1}\cdots \hat{T}_{i_{j_0}}\cdots T_{i_k}(1)) & , \quad i_{j_0} = l \\ f(i(-1)^{|B|}T_lT_{i_1}\cdots T_{i_k}(1)) & , \quad 1 \leq j \leq k, i_j \neq l \end{cases} \\
 &= i \begin{cases} (-1)^{j_0}e_{2i_1-1} \wedge e_{2i_2-1} \wedge \cdots \wedge \hat{e}_{2i_{j_0}-1} \wedge \cdots \wedge e_{2i_k-1} , & i_{j_0} = l \\ e_{2l-1} \wedge e_{2i_1-1} \wedge \cdots \wedge e_{2i_k-1} , & 1 \leq j \leq k, \quad l \leq i_j \\ (-1)^{j_1}e_{2i_1-1} \wedge \cdots \wedge e_{2i_{j_1}-1} \wedge e_{2l-1} \wedge e_{2i_{j_2}-1} \\ \quad \wedge \cdots \wedge e_{2i_k-1} , & i_{j_1} < l < i_{j_2} \\ (-1)^k e_{2i_1-1} \wedge e_{2i_2-1} \wedge \cdots \wedge e_{2i_k-1} \wedge e_{2l-1} , & 1 \leq j \leq k, \quad l \geq i_j. \end{cases} \quad (3.5)
 \end{aligned}$$

On the other hand,

$$\begin{aligned}
 \pi_J(e_{2l})[f(T_{i_1}T_{i_2}\cdots T_{i_k}(1))] &= \pi_J(e_{2l})[e_{2i_1-1} \wedge e_{2i_2-1} \wedge \cdots \wedge e_{2i_k-1}] \\
 &= \begin{cases} -i(-1)^{j_0-1}e_{2i_1-1} \wedge e_{2i_2-1} \wedge \cdots \wedge \hat{e}_{2i_{j_0}-1} \wedge \cdots \wedge e_{2i_k-1} , & i_{j_0} = l \\ ie_{2l-1} \wedge e_{2i_1-1} \wedge \cdots \wedge e_{2i_k-1} , & 1 \leq j \leq k, \quad l \leq i_j \\ i(-1)^{j_1}e_{2i_1-1} \wedge \cdots \wedge e_{2i_{j_1}-1} \wedge e_{2l-1} \wedge e_{2i_{j_2}-1} \\ \quad \wedge \cdots \wedge e_{2i_k-1} , & i_{j_1} < l < i_{j_2} \\ i(-1)^k e_{2i_1-1} \wedge e_{2i_2-1} \wedge \cdots \wedge e_{2i_k-1} \wedge e_{2l-1} , & 1 \leq j \leq k, \quad l \geq i_j. \end{cases} \quad (3.6)
 \end{aligned}$$

Thus we get the equality,

$$(f \circ \psi(e_{2l}))(T_{i_1}T_{i_2}\cdots T_{i_k}(1)) = (\pi_J(e_{2l}) \circ f)(T_{i_1}T_{i_2}\cdots T_{i_k}(1))$$

from the equalities (3.5) and (3.6). This completes the proof.  $\square$

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